UNDER CONDITION V THE PULLBACK ON COHOMOLOGY IS INJECTIVE WITH IMAGE THE CLASSES TRIVIALISED BY A COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reformulation of the map through epsilon_X, the exact sequence obtained from the Leray spectral sequence, and the reading off of injectivity and the image.

Editorial note. The label lemma-V-implies-cohomology-extra-general names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to the standing situation of the comparison of two topologies and is delivered verbatim; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The standing situation comparing two topologies and the functors epsilon_X, epsilon_X^{-1}. Same chapter, the situation comparing the topologies tau and tau' together with the subcategories A_X and the quadratic identity epsilon_X^{-1} epsilon_{X,*} = id on A_X. Supplies the objects and the description of the map as pullback of cohomology classes.
- The Leray spectral sequence for the morphism epsilon_X. Same chapter, the Leray spectral sequence E_2^{p,q} = H^p_{tau'}(X, R^q epsilon_{X,*}F) converging to H^{p+q}_tau(X, F). Supplies the exact sequence from which the statement is read off.

Source reference. Original reference 01F2: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2323–2323. Prerequisite pointer invoked at proof line(s) 7073; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6841–6841. Prerequisite pointer invoked at proof line(s) 7062; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 0EZ3 if (V_n) holds, then for X in $\mathcal{C}$ and $\mathcal{F}$ in $\mathcal{A}_X$ the map $H^{n+1}_{\tau'}(X, \epsilon_{X,*}\mathcal{F}) \to H^{n+1}_\tau(X, \mathcal{F})$ is injective with image those classes which become trivial on a τ'-covering of X.*

Proof. Recall that $\epsilon_X^{-1}\epsilon_{X,*}\mathcal{F} = \mathcal{F}$ hence the map is given by pulling back cohomology classes by ϵ_X. The Leray spectral sequence (Lemma 01F2)

$$E_2^{p,q} = H^p_{\tau'}(X, R^q\epsilon_{X,*}\mathcal{F}) \Rightarrow H^{p+q}_\tau(X, \mathcal{F})$$

combined with the assumed vanishing gives an exact sequence

$$0 \to H^{n+1}_{\tau'}(X, \epsilon_{X,*}\mathcal{F}) \to H^{n+1}_\tau(X, \mathcal{F}) \to H^0_{\tau'}(X, R^{n+1}\epsilon_{X,*}\mathcal{F})$$

This is a restatement of the lemma. □